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Genetics and Evolution

Edexcel A A-Level Biology — Topic 5

5.1 DNA Structure and Replication

DNA (deoxyribonucleic acid): A double-stranded polynucleotide that carries the hereditary genetic information in all living organisms (except some viruses which use RNA). Each nucleotide consists of a deoxyribose sugar (5-carbon pentose), a phosphate group, and one of four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G). Purines (A, G) have a double-ring structure; pyrimidines (T, C) have a single-ring structure.

DNA Structure

- Two antiparallel polynucleotide strands form a **double helix**, first described by Watson and Crick (1953) using X-ray crystallography data from Franklin and Wilkins
- **Sugar-phosphate backbone** on the outside — nucleotides linked by covalent **phosphodiester bonds** (between the 3' carbon of one sugar and the 5' phosphate of the next). This gives the backbone great stability.
- Bases point inward and pair by **complementary base pairing**: A–T (2 hydrogen bonds), C–G (3 hydrogen bonds). This means a purine always pairs with a pyrimidine, keeping the helix a constant width.
- Strands are **antiparallel**: one runs 5'→3', the other 3'→5'. This has implications for replication (leading vs lagging strand).
- The double helix makes one complete turn every 10 base pairs (~3.4 nm per turn), with a diameter of ~2 nm

Why DNA is Suited to Its Function

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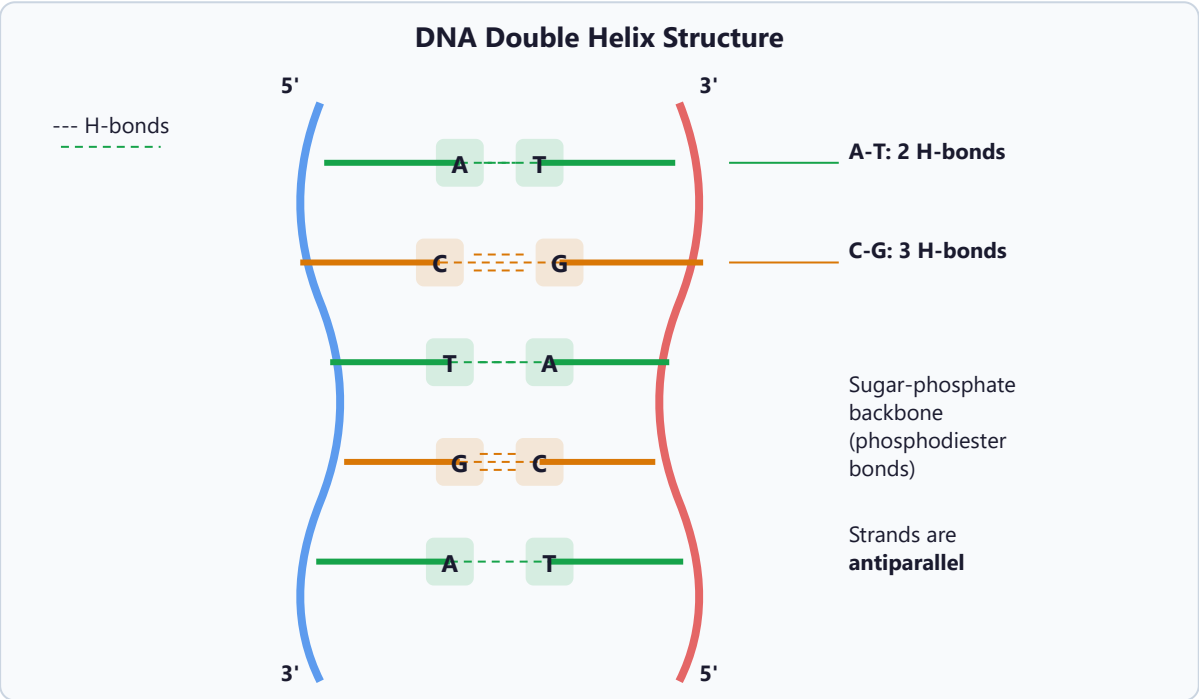
- **Stable** — phosphodiester bonds in backbone are strong covalent bonds; double helix is reinforced by many hydrogen bonds and base stacking interactions
- **Base pairing allows accurate replication** — each strand serves as a template for producing a complementary copy
- **Very long molecule** — can carry a vast amount of genetic information (the human genome has ~3.2 billion base pairs)
- **Two separate strands** can unwind for replication and transcription, then re-form
- **Base sequence** carries the genetic code — infinite variety of sequences possible



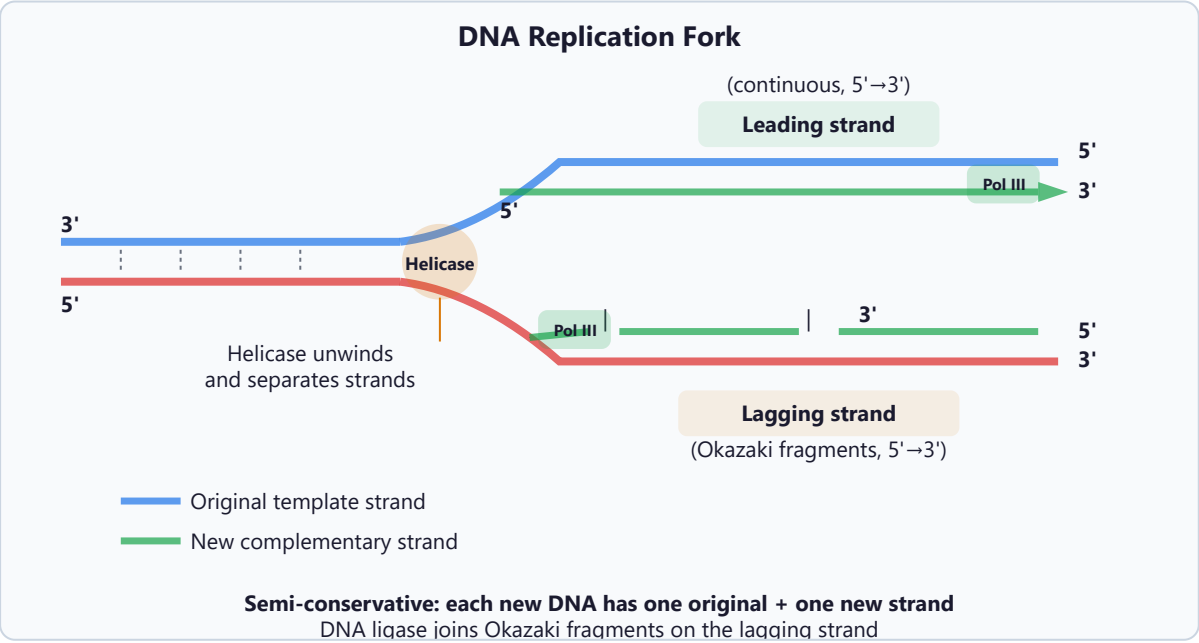
Chargaff's Rules

Erwin Chargaff discovered that in any DNA sample: the amount of adenine equals thymine (A = T), and the amount of cytosine equals guanine (C = G). This provided crucial evidence for complementary base pairing. Therefore: %A + %T + %C + %G = 100%, and if %A = 30%, then %T = 30%, leaving %C = %G = 20% each.

Semi-Conservative Replication

DNA replication is described as semi-conservative because each new DNA molecule consists of one original (conserved) strand and one newly synthesised strand.

1. **Helicase** unwinds the double helix by breaking hydrogen bonds between complementary base pairs, separating the two strands and creating a **replication fork**. Single-strand binding proteins (SSBs) stabilise the separated strands and prevent them re-annealing.
2. Each exposed strand acts as a **template** for the synthesis of a new complementary strand
3. **DNA primase** synthesises a short RNA primer on each template strand (DNA polymerase cannot initiate synthesis from scratch — it can only extend an existing strand)
4. **DNA polymerase III** adds free DNA nucleotides (deoxyribonucleoside triphosphates) to the 3' end of the primer/growing strand by complementary base pairing (A with T, C with G). The energy for bond formation comes from hydrolysis of the two extra phosphate groups on each nucleotide.
5. DNA polymerase works in the **5'→3' direction only**:
   - **Leading strand** is synthesised continuously in the same direction as fork movement
   - **Lagging strand** is synthesised discontinuously in short **Okazaki fragments** (each requiring its own RNA primer), because polymerase must work away from the fork
6. **DNA polymerase I** removes the RNA primers and replaces them with DNA nucleotides
7. **DNA ligase** joins the Okazaki fragments by forming phosphodiester bonds between adjacent nucleotides, creating a continuous strand
8. Result: two identical DNA molecules, each with one original strand and one new strand (semi-conservative)



Evidence for Semi-Conservative Replication: Meselson-Stahl Experiment (1958)

This elegant experiment used *E. coli* and heavy nitrogen (^{15}N) with density gradient centrifugation:

1. Bacteria grown for many generations in ^{15}N medium — all DNA becomes "heavy" (both strands contain ^{15}N)
2. Transferred to ^{14}N (light) medium and allowed to replicate once
3. **After one generation:** All DNA was **intermediate density** (one heavy strand + one light strand) — this ruled out conservative replication (which would have produced one heavy and one light band)
4. **After two generations:** DNA was 50% intermediate and 50% light — exactly as predicted by semi-conservative replication

Exam Tip: The Meselson-Stahl experiment is a key piece of evidence you must be able to describe. Explain why each result supports semi-conservative and rules out conservative replication. After generation 1: all intermediate density proves each molecule has one old and one new strand. Conservative replication would produce one fully heavy and one fully light molecule instead.

5.2 Protein Synthesis

Genes and the Genetic Code

Gene: A section of DNA that codes for a particular polypeptide (or functional RNA molecule such as tRNA or rRNA). A gene's nucleotide sequence determines the amino acid sequence of the protein it encodes, via the triplet code.

The genetic code: A triplet code (three bases = one codon = one amino acid). It is **degenerate** (most amino acids are coded by more than one codon — 64 possible codons for 20 amino acids + 3 stop codons), **universal** (the same code is used by virtually all organisms,

> providing evidence for a common ancestor), **non-overlapping** (each base is read only once), and read in a fixed reading frame from a start codon (AUG).

Transcription (Nucleus)

Transcription is the synthesis of an mRNA molecule from a DNA template. It produces a single-stranded RNA copy of one gene.

1. **RNA polymerase** binds to the **promoter region** (a specific DNA sequence upstream of the gene) on the **template (antisense) strand**. Transcription factors must bind first to facilitate this.
2. Hydrogen bonds between DNA base pairs are broken; DNA strands separate locally (forming a "transcription bubble")
3. RNA polymerase moves along the template strand in the **3'→5' direction**, assembling a complementary mRNA molecule in the **5'→3' direction**
4. Free RNA nucleotides (ribonucleoside triphosphates) are added by complementary base pairing: C pairs with G, G pairs with C, A pairs with T (in DNA), and **U (uracil) pairs with A** (RNA uses uracil instead of thymine)
5. RNA polymerase reaches a **terminator sequence** and detaches; the mRNA is released. The DNA double helix re-forms behind the polymerase.
6. In eukaryotes: the initial transcript is **pre-mRNA**, which undergoes **post-transcriptional modification**:
   - **Splicing** — introns (non-coding sequences) are removed and exons (coding sequences) are joined together by spliceosomes
   - **5' capping** — a modified guanine nucleotide is added to the 5' end (protects from degradation, aids ribosome binding)
   - **3' polyadenylation** — a poly-A tail (~200 adenine nucleotides) is added to the 3' end (protects from exonuclease degradation, aids nuclear export)
7. Mature mRNA leaves the nucleus through **nuclear pores** and enters the cytoplasm for translation

Translation (Ribosomes in Cytoplasm)

Translation is the synthesis of a polypeptide from an mRNA template. It takes place on ribosomes (which may be free in the cytoplasm or bound to the rough endoplasmic reticulum).

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1. mRNA binds to the **small ribosomal subunit**; the start codon (**AUG**) is positioned at the **P site** (peptidyl site)
2. **tRNA** (transfer RNA) molecules are adaptor molecules — each carries a specific amino acid and has an **anticodon** (three bases) complementary to a specific mRNA codon. Aminoacyl-tRNA synthetases attach the correct amino acid to each tRNA.
3. The first tRNA (carrying methionine, with anticodon UAC) binds to the start codon AUG at the P site by complementary base pairing between anticodon and codon
4. A second tRNA (carrying the next amino acid) binds at the **A site** (aminoacyl site) — its anticodon matches the next mRNA codon
5. A **peptide bond** forms between the amino acids on adjacent tRNAs (catalysed by the ribosome itself — specifically by rRNA, making it a ribozyme)
6. The ribosome **translocates** one codon along the mRNA in the 5'→3' direction; the first tRNA (now empty) moves to the **E site** (exit site) and is released; the second tRNA (now carrying the dipeptide) moves to the P site; the A site is free for the next tRNA
7. Process repeats — the polypeptide chain elongates one amino acid at a time
8. When a **stop codon** (UAA, UAG, or UGA) enters the A site, no tRNA can bind. A **release factor** protein binds instead, causing the ribosome to release the polypeptide chain and dissociate from the mRNA.
9. The polypeptide folds into its functional protein shape (primary → secondary → tertiary → quaternary structure). Post-translational modifications may occur (e.g., glycosylation in the Golgi).

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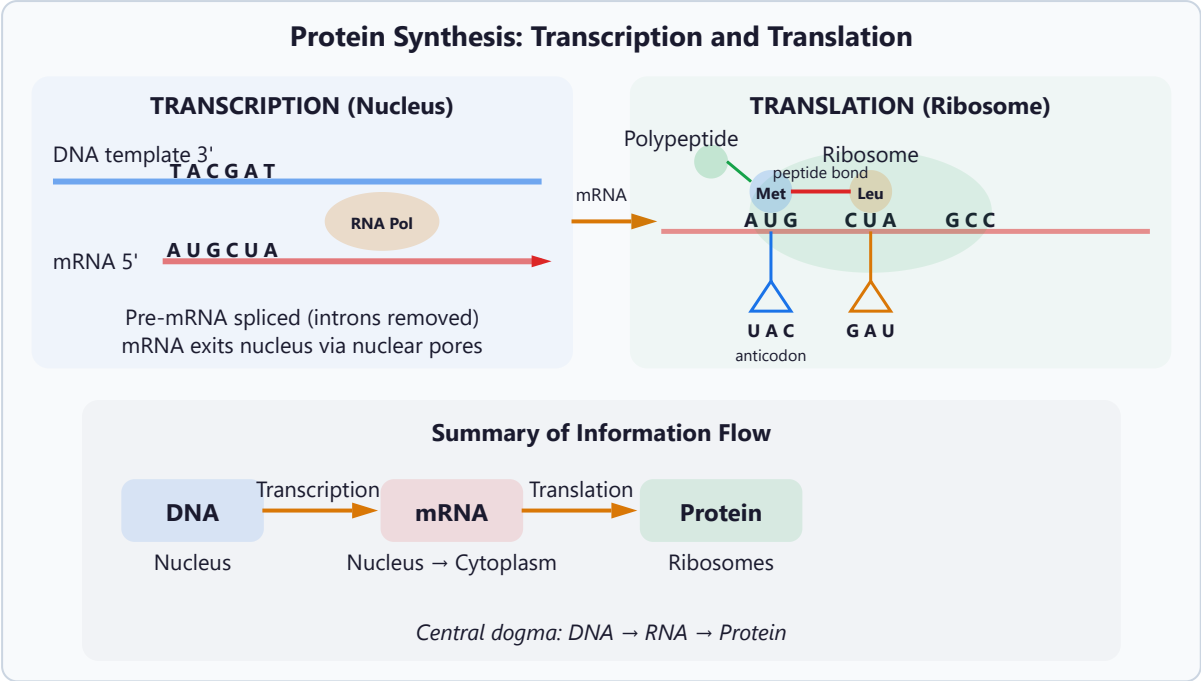

5.3 Gene Mutations

Gene mutation: A change in the nucleotide sequence of DNA. Mutations are the ultimate source of all genetic variation and can be spontaneous (arising from errors during DNA replication, ~1 in 10^9 bases per cell division) or induced by **mutagens**: UV radiation (causes thymine dimers), ionising radiation (causes strand breaks), chemical mutagens (e.g., base analogues, alkylating agents, deaminating agents), and some viruses.

Types of Point Mutation

Type	Change	Effect on Protein	Example
Substitution (silent/synonymous)	One base replaced by another	No change to amino acid sequence — degenerate code means same amino acid coded (often in the third position of the codon, the "wobble" position)	CUU → CUC (both code for leucine)

Substitution (missense)	One base replaced	Different amino acid incorporated — may or may not alter protein function depending on the chemical similarity of the amino acids and the location in the protein	Sickle cell: GAG → GTG (Glu → Val in haemoglobin beta chain)
Substitution (nonsense)	One base replaced	Creates a premature stop codon — translation terminates early; truncated, usually non-functional protein	CAG (Gln) → UAG (stop)
Insertion	One or more bases added	**Frameshift** — all codons downstream of the insertion are altered; usually produces a completely non-functional protein or premature stop codon	Addition of a base shifts the reading frame
Deletion	One or more bases removed	**Frameshift** — same devastating effect as insertion; all downstream codons affected	Cystic fibrosis (most common mutation is deletion of 3 bases = loss of one amino acid, not a frameshift but still pathogenic)

Exam Tip: Frameshift mutations (insertions and deletions of non-multiples of three) are generally MORE damaging than substitutions because they alter EVERY codon after the mutation point, producing a completely different amino acid sequence. A substitution typically only affects one amino acid. If three bases are inserted or deleted together, it adds or removes one amino acid without shifting the frame — still harmful but less catastrophic. The most common CF mutation (ΔF508) is a 3-base deletion that removes phenylalanine at position 508 but does not cause a frameshift.

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Chromosome Mutations

Larger-scale mutations affecting whole chromosomes or large segments:

- **Aneuploidy** — gain or loss of individual chromosomes due to non-disjunction during meiosis. Example: Down syndrome (trisomy 21, three copies of chromosome 21)
- **Polyploidy** — having more than two complete sets of chromosomes (3n, 4n, etc.). Common in plants; can lead to instant speciation (sympatric). Many crop plants are polyploid (e.g., wheat is hexaploid, 6n).
- **Translocation** — segment of one chromosome breaks off and joins a different chromosome
- **Inversion** — segment of chromosome is reversed in orientation
- **Duplication / deletion** — segments are duplicated or lost

5.4 Inheritance

Key Terminology

Term	Definition
Gene	A section of DNA that codes for a particular polypeptide or functional RNA molecule
Allele	An alternative form of a gene; different alleles arise by mutation and have different nucleotide sequences at the same locus
Locus	The specific, fixed position of a gene on a chromosome
Genotype	The combination of alleles an organism possesses for a particular gene (or set of genes)
Phenotype	The observable characteristics of an organism, resulting from the interaction of genotype and environment
Homozygous	Two identical alleles at a locus (e.g., AA or aa)
Heterozygous	Two different alleles at a locus (e.g., Aa)

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Dominant	Allele that is expressed in both homozygous (AA) and heterozygous (Aa) individuals; the dominant phenotype is produced when at least one copy is present
Recessive	Allele only expressed in the homozygous condition (aa); masked by the dominant allele in heterozygotes
Codominant	Both alleles are expressed equally in the heterozygote; neither is dominant over the other
Multiple alleles	More than two alleles exist for a gene in the population (though each individual still has only two)
Test cross	Crossing an individual showing the dominant phenotype with a homozygous recessive individual to determine the unknown genotype

Monohybrid Inheritance

Involves one gene with two alleles. Use Punnett squares to predict offspring ratios. A cross between two heterozygotes (Aa × Aa) gives a **3:1 phenotypic ratio** (3 dominant : 1 recessive) and a 1:2:1 genotypic ratio (1 AA : 2 Aa : 1 aa). These are expected (theoretical) ratios; actual results may deviate due to chance, especially with small sample sizes. A chi-squared test can be used to determine whether observed ratios differ significantly from expected.

Dihybrid Inheritance

Involves two genes on different chromosomes (independently assorting according to Mendel's law of independent assortment). A cross between two double heterozygotes (AaBb × AaBb) gives a **9:3:3:1 phenotypic ratio**: 9 with both dominant traits, 3 with first dominant and second recessive, 3 with first recessive and second dominant, 1 with both recessive.

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Punnett Squares: Monohybrid and Dihybrid Crosses

Monohybrid: Aa x Aa

Gametes	A	a
A	AA	Aa
a	Aa	aa

Genotypic: 1 AA : 2 Aa : 1 aa
Phenotypic: 3 dominant : 1 recessive

Dihybrid: AaBb x AaBb

	AB	Ab	aB	ab
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	AAbb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	aabb

Phenotypic ratio: 9 : 3 : 3 : 1

- 9 A_B_ (both dominant)
- 3 A_bb (A dom, B rec)
- 3 aaB_ (A rec, B dom)
- 1 aabb (both recessive)

Dihybrid 9:3:3:1 ratio only applies with independent assortment (genes on different chromosomes)

Codominance and Multiple Alleles

Codominance: Both alleles are equally expressed in the heterozygote (neither is dominant). The heterozygote shows a phenotype distinct from either homozygote, expressing both allelic products simultaneously.

ABO blood groups demonstrate both codominance and multiple alleles:

- The ABO gene has three alleles: I^A, I^B, and i
- I^A and I^B are **codominant** with each other (I^AI^B genotype gives blood group AB, expressing both A and B antigens on red blood cells)
- Both I^A and I^B are **dominant** over i (recessive)
- Blood group O = ii (no A or B antigens); group A = I^AI^A or I^Ai; group B = I^BI^B or I^Bi; group AB = I^AI^B

Sex Linkage

Genes located on the **X chromosome** (X-linked genes) show a characteristic pattern of inheritance. Males (XY) are **hemizygous** for X-linked genes — they have only one allele, so a single recessive allele will be expressed. This is why X-linked recessive conditions are much more common in males.

Examples: haemophilia A (factor VIII deficiency), red-green colour blindness, Duchenne muscular dystrophy.

Carrier females (X^HX^h) are phenotypically normal but can pass the recessive allele to their sons. An affected father (X^hY) cannot pass the condition to his sons (he gives them his Y chromosome) but all his daughters will be carriers.

Autosomal Linkage

Genes located on the **same chromosome** tend to be inherited together and do not independently assort (violating Mendel's law of independent assortment). Linked genes produce fewer recombinant offspring than the expected 9:3:3:1 ratio. **Crossing over** during meiosis I can break linkage, producing recombinant gametes. The frequency of recombination is proportional to the distance between genes on the chromosome — this principle is used to construct **genetic maps**.

Epistasis

Epistasis occurs when one gene masks or modifies the expression of another gene at a different locus. This produces modified dihybrid ratios. The masking gene is said to be "epistatic to" the masked gene.

Type	Modified Ratio	Example
Recessive epistasis	9:3:4	Coat colour in Labrador retrievers: the ee genotype at the E locus masks the B/b locus, producing yellow regardless of B genotype
Dominant epistasis	12:3:1	Squash fruit colour: a dominant allele at one locus masks expression of another locus
Complementary genes	9:7	Flower colour in sweet peas: two genes both needed for pigment production; homozygous recessive at either locus = white
Duplicate genes	15:1	Kernel colour: a dominant allele at either locus produces the same phenotype

Exam Tip: In genetics problems: (1) Define your allele symbols clearly (use a key), (2) State the parental genotypes and phenotypes, (3) Show the gametes each parent can produce, (4) Draw

a complete Punnett square, (5) State offspring genotypic and phenotypic ratios. For sex linkage, always use X^HX^h notation showing the alleles as superscripts on the X chromosome. For dihybrid crosses, check whether the observed ratio matches 9:3:3:1 or a modified epistatic ratio.

Chi-Squared Test

The chi-squared (χ^2) test determines whether the difference between observed and expected results is statistically significant or due to chance alone.

$$\chi^2 = \Sigma (O - E)^2 / E$$

Where O = observed frequency, E = expected frequency. Sum across all categories.

1. State the **null hypothesis** (H_0): there is no significant difference between observed and expected results
2. Calculate the expected values from the predicted ratio and total number of offspring
3. Calculate χ^2 using the formula
4. Determine **degrees of freedom** = number of categories − 1
5. Compare χ^2 with the critical value at $p = 0.05$ (5% significance level)
6. If χ^2 < critical value: accept H_0 (difference is due to chance; results fit the expected ratio)
7. If χ^2 > critical value: reject H_0 (difference is statistically significant; results do not fit the expected ratio)

5.5 Hardy-Weinberg Principle

Hardy-Weinberg principle: In a large, randomly mating population with no selection, mutation, migration, or genetic drift, allele frequencies and genotype frequencies remain constant from generation to generation. This provides a mathematical model against which real populations can be compared to detect evolutionary change.

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p + q = 1   (where p = frequency of dominant allele, q = frequency of
recessive allele)
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$p^2 + 2pq + q^2 = 1$ (p^2 = freq of AA, $2pq$ = freq of Aa, q^2 = freq of aa)

Using Hardy-Weinberg in Calculations

1. Identify the number or frequency of **homozygous recessive individuals** (q^2) — these are the only genotype identifiable directly from phenotype in a dominant/recessive system
2. Take the square root to find q (frequency of recessive allele)
3. Calculate $p = 1 - q$ (frequency of dominant allele)
4. Calculate carrier (heterozygote) frequency: $2pq$
5. Calculate homozygous dominant frequency: p^2

Conditions for Hardy-Weinberg equilibrium (all must be met — they rarely are in nature, which is why evolution occurs):

- Large population size (no **genetic drift**)
- Random mating (no **sexual selection** or assortative mating)
- No **natural selection** (all genotypes equally fit)
- No **mutation** (no new alleles introduced)
- No **migration** (no gene flow into or out of the population)

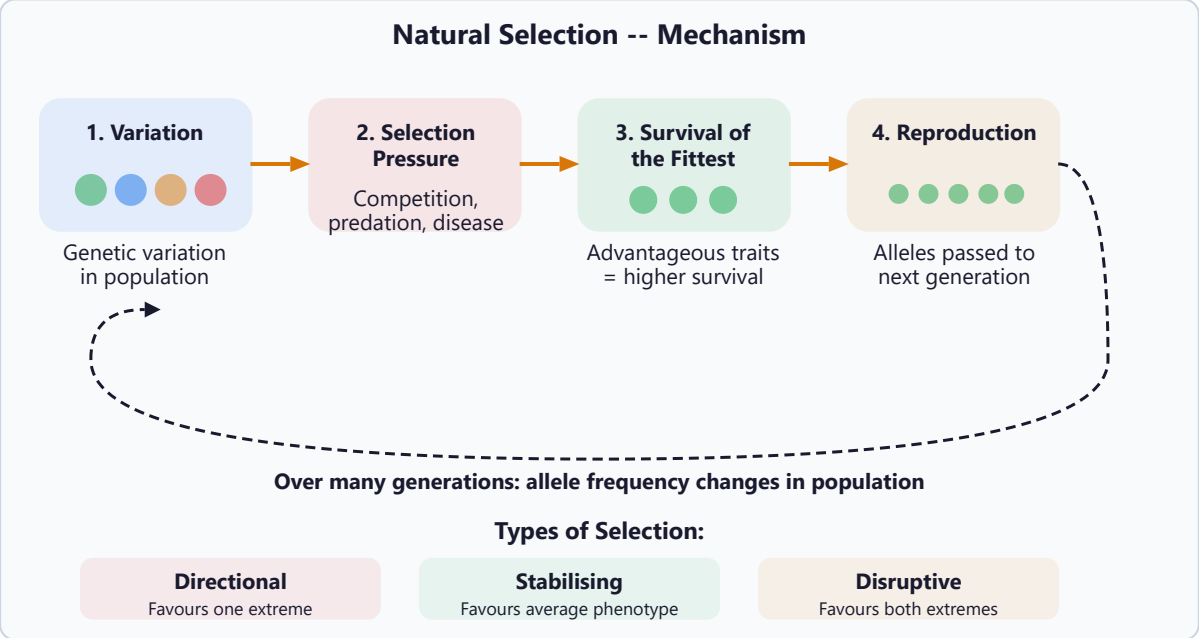
Exam Tip: In Hardy-Weinberg calculations, always start with q^2 (the proportion of homozygous recessive individuals). You cannot start with p^2 because you cannot distinguish AA from Aa by phenotype alone in a dominant/recessive system. Always show your working clearly: state what q^2 is, calculate q, then p, then 2pq if needed.

5.6 Natural Selection and Evolution

Evolution: A change in allele frequency in a population over successive generations. It is driven by four main mechanisms: natural selection, genetic drift, mutation, and gene flow (migration). Natural selection is the only mechanism that is adaptive (leads to populations becoming better suited to their environment).

Mechanism of Natural Selection

1. **Variation** exists within the population due to genetic differences (caused by mutation, meiosis, and sexual reproduction)
2. There is a **selection pressure** (environmental factor such as competition for resources, predation, disease, or climate change)
3. Individuals with advantageous phenotypes have a **selective advantage** — they are more likely to survive to reproductive age ("survival of the fittest")
4. These individuals are more likely to **reproduce** and pass their advantageous alleles to the next generation
5. Over many generations, the **frequency of the advantageous allele increases** in the population
6. The population becomes better adapted to its environment



Types of Natural Selection

Type	What It Favours	Effect on Distribution	Example
Directional	One phenotypic extreme	Mean shifts towards the favoured extreme; distribution shifts left or right	Antibiotic resistance in bacteria (resistant individuals survive); peppered moth (industrial melanism — dark moths favoured in polluted environments)
Stabilising	Average/intermediate phenotype	Reduces variation; narrows the distribution; extremes are selected against	Human birth weight (very small or very large babies have lower survival); clutch size in birds
Disruptive	Both phenotypic extremes	Increases variation; creates a bimodal distribution; may lead to speciation	Beak size in African seedcrackers (large beaks for hard seeds, small for soft; intermediate beaks disadvantaged)

Genetic Drift

Random changes in allele frequency due to chance events in reproduction, particularly significant in **small populations** where sampling error is large. Drift is non-adaptive (does not lead to better adaptation) and can lead to **fixation** (allele frequency reaches 1.0) or **loss** (frequency reaches 0) regardless of whether the allele is advantageous. Two important special cases:

- **Bottleneck effect:** A dramatic, often sudden, reduction in population size (natural disaster, disease, habitat destruction) randomly eliminates alleles. The surviving population has reduced genetic diversity and may have different allele frequencies from the original.
- **Founder effect:** A small group of individuals colonises a new area (e.g., island). Their limited gene pool differs from the parent population by chance, and rare alleles may be overrepresented. Example: high frequency of Ellis-van Creveld syndrome among Old Order Amish in Pennsylvania, descended from a small founding group.

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5.7 Speciation

Species (biological species concept): A group of organisms that can interbreed to produce fertile offspring and are reproductively isolated from other groups. **Speciation** is the formation of new species, which requires reproductive isolation leading to genetic divergence.

Type	Mechanism	Key Feature	Example
Allopatric speciation	Geographic isolation (physical barrier such as mountain range, river, ocean, glacier) separates populations → different selection pressures in different environments → different mutations accumulate → allele frequencies diverge independently → eventually reproductively isolated = new species	Requires a physical barrier	Darwin's finches on different Galapagos Islands; squirrels on opposite sides of the Grand Canyon
Sympatric speciation	Reproductive isolation develops within the same geographic area, without physical separation. Can occur through polyploidy (instant reproductive isolation in plants), ecological isolation (using different resources or microhabitats), temporal isolation (breeding at different times), or behavioural isolation (different mating calls/displays).	No physical barrier	Polyploidy in plants (e.g., bread wheat arose from hybridisation and chromosome doubling); cichlid fish in African lakes diverging by mate preference

Reproductive Isolation Mechanisms

- **Pre-zygotic barriers** (prevent fertilisation):
  - Temporal/seasonal — different breeding times or seasons
  - Mechanical — incompatible genitalia or flower structures

  - Behavioural — different courtship displays, mating calls, or pheromones (not recognised as potential mates)
  - Gametic — gametes cannot fuse (different surface proteins on sperm/egg)
  - Habitat/ecological — occupy different microhabitats within same area
- **Post-zygotic barriers** (act after fertilisation):
  - Hybrid inviability — hybrid embryo does not develop or dies before reproduction
  - Hybrid sterility — hybrid offspring are viable but infertile (e.g., mules from horse × donkey; ligers from lion × tiger)
  - Hybrid breakdown — first-generation hybrids are fertile but subsequent generations have reduced fitness

> **Exam Tip:** When explaining speciation in an extended answer, follow this logical sequence: (1) populations are separated/isolated (describe how), (2) different selection pressures act in each environment, (3) different random mutations arise independently in each population, (4) natural selection favours different alleles in each population, (5) allele frequencies diverge over many generations, (6) genetic differences accumulate until the populations are reproductively isolated (they can no longer interbreed to produce fertile offspring) = they have become separate species. Always use the term "allele frequency" and link to natural selection.

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